On the Extended, Generalized, and Grand Riemann Hypotheses: A Unified Approach via the Divisibility Property of Entire Functions

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Abstract

This paper addresses the Extended, Generalized, and Grand Riemann Hypotheses within a unified framework via the divisibility property of entire functions (see Lemma 2), based on general properties of L-functions (see P1-P6). First, pairing complex-conjugate zeros in the Hadamard products of the completed L-functions $\Lambda(\lambda, s)$ (for the self-dual case) or $\Lambda(\lambda, s)\Lambda(\bar{\lambda}, s)$ (for the non-self-dual case), together with the fact that all zeros lie in the critical strip, as guaranteed by the Euler product property, yields absolutely and uniformly convergent infinite products with irreducible real quadratic polynomial factors. Second, substituting these infinite products into the symmetric functional equation yields the divisibility property of the completed L-functions. Third, this divisibility property, combined with the uniqueness of zero multiplicities, forces all zeros of $\Lambda(\lambda, s)$ or $\Lambda(\lambda, s)\Lambda(\bar{\lambda}, s)$ in the critical

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strip to lie on the critical line. Consequently, the existence of Landau–Siegel zeros is excluded, thereby confirming the Landau–Siegel zeros conjecture.

As for the Davenport–Heilbronn counterexample, since it possesses no Euler product—the fundamental structural property that confines zeros to the critical strip, thereby guaranteeing the absolute and uniform convergence of the infinite products and, in turn, the divisibility property of entire functions—it falls outside the scope of the methods and conclusions of this paper.

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Keywords: Extended Riemann Hypothesis, Generalized Riemann Hypothesis, Grand Riemann Hypothesis, Landau–Siegel zeros conjecture, Davenport–Heilbronn counterexample, divisibility property of entire functions, uniqueness of zero multiplicities, symmetric functional equation, Hadamard product, Euler product.

1 Notation and Sketch of the Contents

Notation.

$\mathbb{N}$: the set of natural numbers, i.e., $\mathbb{N} = \{1, 2, 3, \ldots\}$.
$\mathbb{R}$: the field of real numbers.
$\mathbb{C}$: the field of complex numbers (the complex plane).
$\mathbb{Q}$, the field of rational numbers.
$\mathbb{R}[x]$: the ring of real polynomials.

It should be noted that in this paper, 'j' is used to denote the imaginary unit ($j^2 = -1$), while 'i' together with 'l, n, t' serves as natural number indices.

It is also worth noting that the following fact (supported by Lemma 4 and Lemma 5) plays an important role in this paper: **for a given non-zero entire function, the multiplicities of its zeros are finite and uniquely determined, hence invariant, even though their specific values may be unknown.**

Sketch of the Contents.

Preliminary Lemmas: Five Lemmas are provided to support the content that follows.
Foundational Theorems: Based on Lemma 2, four theorems are progressively proved from Theorem 1 to Theorem 4.
Main Results: Based on Theorem 4, the generalized Riemann Hypothesis (for Dirichlet L-functions) is proved in Theorem 5; the extended Riemann Hypothesis (for Dedekind zeta functions) is proved in Theorem 6; Theorem 7 deals with the non-trivial zero distribution of cuspidal automorphic L-functions, which corresponds to grand Riemann Hypothesis (version 1); Theorem 8 deals with the non-trivial zero distribution of all kinds of L-functions, which corresponds to grand Riemann Hypothesis (version 2).

2 Preliminary Lemmas

Lemma 1: Let $m(x), g_1(x), ..., g_n(x) \in \mathbb{R}[x], n \geq 2$. If $m(x)$ is irreducible (prime) and divides the product $g_1(x) \cdots g_n(x)$, then $m(x)$ divides one of the polynomials $g_1(x), ..., g_n(x)$.

Remark: The contents of Lemma 1 can be found in many textbooks of linear algebra, modern algebra, or abstract algebra, such as Ref.[1, p.135].

Next, we extends the above polynomial divisibility property in Lemma 1 to entire functions expressed as absolutely and uniformly convergent infinite products with irreducible real polynomial factors.

Lemma 2: Let $f(s)$ be an entire function on $\mathbb{C}$ with $f(0) \neq 0$. Suppose $f(s) = \prod_{i=1}^{\infty} g_i(s)$ converges absolutely and uniformly on compact subsets of $\mathbb{C}$, where each $g_i(s) \in \mathbb{R}[s]$ is irreducible of degree $d \in \{1, 2\}$. If $m(s) \in \mathbb{R}[s]$ is irreducible of degree d and divides $f(s)$, then $m(s)$ divides $g_i(s)$ for some $i \geq 1$.

Proof: Let α be a root of $m(s)$, i.e., $m(\alpha) = 0$. Since $m(s) \mid f(s)$, we have $f(\alpha) = 0$. By absolute convergence of $\prod_{i=1}^{\infty} g_i(s)$ at $s = \alpha$, there exists at least one index $i \in \mathbb{N}$ such that $g_i(\alpha) = 0$, otherwise $g_i(\alpha) \neq 0$ for all i, then $\prod_{i=1}^{\infty} g_i(\alpha)$ converges to a non-zero limit, contradicting $f(\alpha) = 0$.

As $g_i(s)$ and $m(s)$ are irreducible in $\mathbb{R}[s]$ with $\deg(g_i(s)) = \deg(m(s)) = d$, they share the root α. Thus:

- If $d = 1$, then $g_i(s) = a(s - \alpha)$ and $m(s) = b(s - \alpha)$ for $a, b \neq 0$, so $m(s) \mid g_i(s)$.

- If $d = 2$, then both $g_i(s)$ and $m(s)$ have roots $\{\alpha, \bar{\alpha}\}$, so $g_i(s) = c \cdot m(s)$ for $c \neq 0$, hence $m(s) \mid g_i(s)$.

In both cases, $m(s)$ divides $g_i(s)$.

That completes the proof of Lemma 2.

Remark: Although the uniform convergence is not used directly in the proof of Lemma 2, it guarantees that the infinite product defines an entire function.

Lemma 3: The infinite product $\prod_{i=1}^{\infty} \left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}$ converges to a non-zero constant, where $0 \leq \alpha_i \leq k, 0 < |\beta_1| \leq |\beta_2| \leq |\beta_3|, \cdots, \sum_{i=1}^{\infty} \frac{1}{\beta_i^2} < \infty$ (with each β_i^2 appearing m_i times), and $1 \leq m_i < \infty$ is the multiplicity of zero $\rho_i = \alpha_i + j\beta_i$ of $\xi(s)$.

Proof: Let $a_i = \frac{\alpha_i^2}{\alpha_i^2+\beta_i^2}$, then $\frac{\beta_i^2}{\alpha_i^2+\beta_i^2} = 1 - a_i$. Since $0 \leq \alpha_i \leq k$ and $\beta_i \neq 0$, we have $0 \leq a_i \leq \frac{k^2}{\beta_i^2}$. By the given condition $\sum_{i=1}^{\infty} \frac{1}{\beta_i^2} < \infty$ (with each β_i^2 appearing m_i times), we have equivalently $\sum_{i=1}^{\infty} \frac{m_i}{\beta_i^2} < \infty$ (with $|\beta_1| < |\beta_2| < \cdots$). Hence $\sum_{i=1}^{\infty} m_i a_i \leq k^2 \sum_{i=1}^{\infty} \frac{m_i}{\beta_i^2} < \infty$, which guarantees that the product $\prod_{i=1}^{\infty}(1 - a_i)^{m_i} = \prod_{i=1}^{\infty} \left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}$ converges to a non-zero constant.

That completes the proof of Lemma 3.

Lemma 4: Let $f(s)$ be a non-zero entire function, and let s_0 be a zero of $f(s)$. Then the multiplicity of s_0 is a finite positive integer.

Proof: Let $f(s) \not\equiv 0, s \in \mathbb{C}$, be an entire function, which means it is holomorphic on the whole complex plane. Suppose $f(s)$ has a zero at $s_0 \in \mathbb{C}$ of multiplicity m, then $f(s) = (s - s_0)^m g(s)$, where $g(s)$ is also an entire function and $g(s_0) \neq 0$.

Assume for contradiction that m is infinite, which implies there exists an accumulation point of zeros in the neighbor of s_0. Then, by Identity Theorem for holomorphic functions, and considering "0" is also an entire function, we have $f(s) \equiv 0, s \in \mathbb{C}$, which contradicts the given condition that $f(s) \not\equiv 0, s \in \mathbb{C}$. Thus, the assumption is false, i.e., m must be a finite positive integer.

That completes the proof of Lemma 4.

Lemma 5: Let $f(s)$ be a non-zero entire function, and let s_0 be a zero of $f(s)$. Then the multiplicity of s_0 is unique.

Proof: Let $f(s) \not\equiv 0, s \in \mathbb{C}$, be an entire function, which has a multiple zero at $s_0 \in \mathbb{C}$ of multiplicity m. We can write: $f(s) = (s - s_0)^m g(s)$, where $g(s)$ is also an entire function and $g(s_0) \neq 0$.

Assume for contradiction that there exists another integer $n \neq m$ such that n is also a multiplicity of the zero s_0. This means we can also write: $f(s) = (s - s_0)^n h(s)$, where $h(s)$ is an entire function and $h(s_0) \neq 0$.

Since both expressions for $f(s)$ must be equal, we then obtain $(s - s_0)^m g(s) = (s - s_0)^n h(s)$. Without loss of generality, consider $m > n$, then we have: $(s - s_0)^{m-n} g(s) = h(s) \Rightarrow h(s_0) = 0$, which is a contradiction to $h(s_0) \neq 0$. Thus, the assumption is false, i.e., the multiplicity of a zero of any non-zero entire function is unique.

That completes the proof of Lemma 5.

3 Foundational Theorems

As pointed out in Ref.[2] (p.57), the non-trivial zeros of the zeta function can be enumerated in order of increasing absolute value of their imaginary parts; where zeros whose imaginary parts have the same absolute value are arranged arbitrarily. Consequently, the same enumeration rule applies to the zeros of entire functions in the following contents. We therefore drop the default assumption $|\beta_1| \leq |\beta_2| \leq \cdots$ as a condition hereafter for simplicity.

Theorem 1: Let $F(s)$ be an entire function satisfying $F(s) = F(k - s)$, where $k > 0$ is a real constant, and admitting the following representation, which converges absolutely and uniformly on compact subsets of $\mathbb{C}$:

$$F(s) = \prod_{i=1}^{\infty} \left(1 + \frac{(s - \alpha_i)^2}{\beta_i^2}\right)^{m_i}, \quad s \in \mathbb{C} \tag{1}$$

Here $m_i \geq 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i = \alpha_i + j\beta_i, \bar{\rho}_i, k - \rho_i, k - \bar{\rho}_i)$ of $F(s)$, $0 \leq \alpha_i \leq k$, $\beta_i \neq 0$, $\sum_{i=1}^{\infty} \frac{m_i}{\beta_i^2} < \infty$.

Then

$$\alpha_i = \frac{k}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots \tag{2}$$

Proof. Considering

$$\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i}=\underbrace{\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}}$$

we have from $F(s)=F(k-s)$ and Eq.(1)

$$\prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}}=\prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} \tag{3}$$

where the ith factor appears m_i times in any order.

Based on the conditions $0\leq\alpha_i\leq k$, $\sum_{i=1}^{\infty}\frac{m_i}{\beta_i^2}<\infty$ and the Weierstrass M-Test, both sides of Eq.(3) are infinite products absolutely and uniformly convergent on compact subsets of $\mathbb{C}$. Hence each side can be rearranged into any single quadratic polynomial factor times the remaining product, which stays absolutely and uniformly convergent on compact subsets of $\mathbb{C}$ and hence defines an entire function. Then by the definition of divisibility of entire functions (see Ref.[3, p.174] or Ref.[4]), Eq.(3) implies that each quadratic polynomial factor on either side divides the infinite product on the opposite side, i.e.,

$$\begin{cases} \left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right) \mid \prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} \\ \left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right) \mid \prod_{i=1}^{\infty}\underbrace{\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)\cdots}_{m_i \text{ times}} \end{cases} \tag{4}$$

where "$\mid$" is the divisible sign, $i\in\mathbb{N}=\{1,2,3,\cdots\}$.

Both polynomials $1+\frac{(s-\alpha_i)^2}{\beta_i^2}$ and $1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}$ have discriminant $\Delta=-4\cdot\frac{1}{\beta_i^2}<0$. Hence they are irreducible in $\mathbb{R}[x]$. By Lemma 2, Eq.(4) yields:

$$\begin{cases} \left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right) \mid \left(1+\frac{(k-s-\alpha_l)^2}{\beta_l^2}\right) \\ \left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right) \mid \left(1+\frac{(s-\alpha_l)^2}{\beta_l^2}\right) \\ i,l\in\mathbb{N} \end{cases} \tag{5}$$

For the special kind of polynomials in Eq.(5), "divisible" means "equal", which can be verified by comparing the like terms in the following equation to get $k' = 1$.

$$\Big(1 + \frac{(s-\alpha_i)^2}{\beta_i^2}\Big) = k'\Big(1 + \frac{(k-s-\alpha_l)^2}{\beta_l^2}\Big), k' \neq 0 \in \mathbb{R} \tag{6}$$

Further, due to the uniqueness of the multiplicity m_i, the only solution to Eq.(5) is

$$\begin{cases} \Big(1 + \frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1 + \frac{(k-s-\alpha_i)^2}{\beta_i^2}\Big) \\ \Big(1 + \frac{(k-s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1 + \frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \end{cases}, i \in \mathbb{N} \tag{7}$$

i.e.,

$$1 + \frac{(s-\alpha_i)^2}{\beta_i^2} = 1 + \frac{(k-s-\alpha_i)^2}{\beta_i^2}, i \in \mathbb{N} \tag{7a}$$

otherwise, duplicated zeros (in quadruple) with $\alpha_i + \alpha_l = k, \beta_i^2 = \beta_l^2, l \neq i$ would be generated to change m_i, as a result of

$$\begin{cases} \Big(1 + \frac{(s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1 + \frac{(k-s-\alpha_l)^2}{\beta_l^2}\Big) \\ \Big(1 + \frac{(k-s-\alpha_i)^2}{\beta_i^2}\Big) \mid \Big(1 + \frac{(s-\alpha_l)^2}{\beta_l^2}\Big) \end{cases}, i, l \in \mathbb{N}, i \neq l \tag{7b}$$

i.e.,

$$\begin{cases} 1 + \frac{(s-\alpha_i)^2}{\beta_i^2} = 1 + \frac{(k-s-\alpha_l)^2}{\beta_l^2} \\ 1 + \frac{(k-s-\alpha_i)^2}{\beta_i^2} = 1 + \frac{(s-\alpha_l)^2}{\beta_l^2} \end{cases}, i, l \in \mathbb{N}, i \neq l \tag{7c}$$

By comparing the like terms in Eq.(7a), we obtain $\alpha_i = \frac{k}{2}, i \in \mathbb{N}$. Further, to ensure the uniqueness of m_i while $\alpha_i = \frac{k}{2}$, we need to restrict the β_i values to be distinct, i.e., $0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots$.
That completes the proof of Theorem 1. □

Remark: Theorem 1 (with $k = 1$) applies to the study of the zero distribution of the Riemann ξ-function. Let $\rho_i = \alpha_i + j\beta_i$, with $\beta_i \neq 0, 0 < \alpha_i < 1$, and let m_i be the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$. The Hadamard product of $\xi(s)$, grouped into conjugate pairs, can be written as

$$\begin{aligned} \xi(s) &= \xi(0) \prod_{i=1}^{\infty} \left[\left(1 - \frac{s}{\rho_i}\right) \left(1 - \frac{s}{\bar{\rho}_i}\right) \right]^{m_i} \\ &= \xi(0) \prod_{i=1}^{\infty} \left[\frac{\beta_i^2}{\alpha_i^2 + \beta_i^2} \left(1 + \frac{(s-\alpha_i)^2}{\beta_i^2}\right) \right]^{m_i} \end{aligned}$$

Since $\xi(s) = \xi(1-s)$, we obtain

$$\prod_{i=1}^{\infty}\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} = \prod_{i=1}^{\infty}\left(1+\frac{(1-s-\alpha_i)^2}{\beta_i^2}\right)^{m_i}$$

where $\sum_{i=1}^{\infty}\frac{m_i}{\beta_i^2} < \infty$.

Theorem 2: Let $G(s)$ be an entire function satisfying $G(s) = G(k-s)$, where $k > 0$ is a real constant, and admitting the following representation, which converges absolutely and uniformly on compact subsets of $\mathbb{C}$:

$$\begin{aligned} G(s) &= \prod_{i=1}^{\infty}\left(1-\frac{s}{\rho_i}\right)^{m_i}\left(1-\frac{s}{\bar{\rho}_i}\right)^{m_i} \\ &= \prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}+\frac{(s-\alpha_i)^2}{\alpha_i^2+\beta_i^2}\right)^{m_i} \\ &= \prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}\prod_{i=1}^{\infty}\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i}, s\in\mathbb{C} \end{aligned} \tag{8}$$

where $\rho_i = \alpha_i + j\beta_i$ and $\bar{\rho}_i = \alpha_i - j\beta_i$ ($\beta_i \neq 0$) are the complex-conjugate zeros of $G(s)$; $m_i \geq 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, k-\rho_i, k-\bar{\rho}_i)$ of $G(s)$ with $0 \leq \alpha_i \leq k$, $\sum_{i=1}^{\infty}\frac{m_i}{|\rho_i|^2} < \infty$.

Then

$$\alpha_i = \frac{k}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots \tag{9}$$

Proof. According to Lemma 3, we have

$$\begin{aligned} &G(s) = G(k-s) \\ &\Leftrightarrow \\ &\prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}\prod_{i=1}^{\infty}\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} = \prod_{i=1}^{\infty}\left(\frac{\beta_i^2}{\alpha_i^2+\beta_i^2}\right)^{m_i}\prod_{i=1}^{\infty}\left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} \\ &\Leftrightarrow \\ &\prod_{i=1}^{\infty}\left(1+\frac{(s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} = \prod_{i=1}^{\infty}\left(1+\frac{(k-s-\alpha_i)^2}{\beta_i^2}\right)^{m_i} \\ &\Leftrightarrow \\ &F(s) = F(k-s) \end{aligned} \tag{10}$$

Next, the conditions of Theorem 2 match those of Theorem 1, considering

$$0 \leq \alpha_i \leq k, \sum_{i=1}^{\infty} \frac{m_i}{|\rho_i|^2} < \infty \Rightarrow 0 \leq \alpha_i \leq k, \sum_{i=1}^{\infty} \frac{m_i}{\beta_i^2} < \infty$$

Then by Theorem 1, we have

$$\alpha_i = \frac{k}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots \tag{11}$$

That completes the proof of Theorem 2. □

In what follows, Theorems 3 and 4 are concerned with the completed L- functions with a self-dual mathematical object λ. Moreover, these two theorems provide a foundation for studying the distribution of zeros of the completed L- functions with a non-self-dual mathematical object λ.

Theorem 3: Let $\Lambda(\lambda, s)$ be an entire function satisfying $\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\lambda, k-s)$, where $k > 0$ is a real constant, and admitting the Hadamard product representation:

$$\Lambda(\lambda, s) = e^{A(\lambda)+B(\lambda)s} \prod_{i=1}^{\infty} \Big(1 - \frac{s}{\rho_i}\Big) e^{\frac{s}{\rho_i}}, s \in \mathbb{C} \tag{12}$$

Here λ denotes a mathematical object, $\varepsilon(\lambda)$ is a complex number of absolute value 1, $\rho_i = \alpha_i + j\beta_i$ and $\bar{\rho}_i = \alpha_i - j\beta_i$ ($\beta_i \neq 0$) are the complex-conjugate zeros of $\Lambda(\lambda, s)$; $m_i \geq 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, k-\rho_i, k-\bar{\rho}_i)$ of $\Lambda(\lambda, s)$ with $0 \leq \alpha_i \leq k$, $\sum_{i=1}^{\infty} \frac{m_i}{|\rho_i|^2} < \infty$.

Then

$$\alpha_i = \frac{k}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots \tag{13}$$

Remark: For more details about $\varepsilon(\lambda)$, see Ref.[5] (p.94).

Proof. First, we have

$$\begin{aligned}
\Lambda(\lambda, s) &= e^{A(\lambda)+B(\lambda)s} \prod_{i=1}^{\infty} \Big(1 - \frac{s}{\rho_i}\Big) e^{\frac{s}{\rho_i}} \\
&= e^{A(\lambda)+B(\lambda)s} \prod_{i=1}^{\infty} \Big(1 - \frac{s}{\rho_i}\Big)\Big(1 - \frac{s}{\bar{\rho}_i}\Big) e^{\frac{2\alpha_i s}{\alpha_i^2+\beta_i^2}} \\
&= e^{A(\lambda)+B(\lambda)s} e^{s\cdot\sum_{i=1}^{\infty} \frac{2\alpha_i}{\alpha_i^2+\beta_i^2}} \prod_{i=1}^{\infty} \Big(1 - \frac{s}{\rho_i}\Big)\Big(1 - \frac{s}{\bar{\rho}_i}\Big)
\end{aligned} \tag{14}$$

Noting that $\sum_{i=1}^{\infty}\frac{2\alpha_i}{\alpha_i^2+\beta_i^2} \leq 2k\cdot\sum_{i=1}^{\infty}\frac{1}{|\rho_i|^2} < \infty$, then we have $\sum_{i=1}^{\infty}\frac{2\alpha_i}{\alpha_i^2+\beta_i^2} = c, c \in \mathbb{R}$. Further, taking into account the multiplicity $m_i \geq 1$ of each zero, we rewrite the product as

$$\begin{aligned}\Lambda(\lambda,s) &= e^{A(\lambda)+[B(\lambda)+c]s}\prod_{i=1}^{\infty}\Big(1-\frac{s}{\rho_i}\Big)^{m_i}\Big(1-\frac{s}{\bar{\rho}_i}\Big)^{m_i} \\ &= e^{A(\lambda)+[B(\lambda)+c]s}G(s)\end{aligned} \tag{15}$$

Accordingly

$$\begin{aligned}\Lambda(\lambda,k-s) &= e^{A(\lambda)+[B(\lambda)+c](k-s)}\prod_{i=1}^{\infty}\Big(1-\frac{k-s}{\rho_i}\Big)^{m_i}\Big(1-\frac{k-s}{\bar{\rho}_i}\Big)^{m_i} \\ &= e^{A(\lambda)+[B(\lambda)+c](k-s)}G(k-s)\end{aligned} \tag{16}$$

Because $e^{A(\lambda)+[B(\lambda)+c]s}$ and $\varepsilon(\lambda)e^{A(\lambda)+[B(\lambda)+c](k-s)}$ are units in the ring of entire functions, they have no zeros and therefore do not affect the complex zeros related divisibility in the functional equation $\Lambda(\lambda,s) = \varepsilon(\lambda)\Lambda(\lambda,k-s)$. We thus conclude that Theorem 3 holds by Theorem 2, whose detailed conditions match those of Theorem 3.

In addition to Eq.(13), we still have (by canceling the complex zeros related polynomial factors on both sides of functional equation: $\Lambda(\lambda,s) = \varepsilon(\lambda)\Lambda(\lambda,k-s)$, since we already have $\alpha_i = \frac{k}{2}$)

$$\Lambda(\lambda,s) = \varepsilon(\lambda)\Lambda(\lambda,k-s) \Longrightarrow e^{A(\lambda)+[B(\lambda)+c]s} = \varepsilon(\lambda)e^{A(\lambda)+[B(\lambda)+c](k-s)} \tag{17}$$

That completes the proof of Theorem 3.

□

In the following Theorem 4, we suppose that the completed L-functions possess both real and complex zeros, denoted by $\rho \in \mathcal{Z}_r$ ($\Im(\rho) = 0$) and $\rho \in \mathcal{Z}_c$ ($\Im(\rho) \neq 0$), respectively. When these two zero sets have no common elements, we express their disjointness by: $\mathcal{Z}_r \cap \mathcal{Z}_c = \varnothing$, which is automatically satisfied by our definitions of $\mathcal{Z}_r$ and $\mathcal{Z}_c$ because $\mathcal{Z}_r$ and $\mathcal{Z}_c$ are mutually exclusive sets, i.e., if $\rho \in \mathcal{Z}_r$, then $\Im(\rho) = 0$; if $\rho \in \mathcal{Z}_c$, then $\Im(\rho) \neq 0$.

The reason we need to consider this case is that, so far, we cannot rule out the existence of exceptional zeros (or Landau-Siegel zeros), although their numbers are very limited even if they do exist.

Denote the set of real zeros as

$$\mathcal{Z}_r = \{a_n \in \mathbb{R} \mid 0 < a_n < k,\, n = 1, 2, \ldots, N\} \tag{18}$$

where N is a finite natural number. This finiteness follows from the Identity Theorem, which implies that any non-zero entire function cannot have infinitely many zeros in a bounded region.

Theorem 4: Let $\Lambda(\lambda, s)$ be an entire function satisfying $\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\lambda, k-s)$, where $k > 0$ is a real constant, and admitting the Hadamard product representation:

$$\begin{aligned}
\Lambda(\lambda, s) &= e^{A(\lambda)+B(\lambda)s} \prod_{\rho} \Big(1 - \frac{s}{\rho}\Big) e^{\frac{s}{\rho}}, s \in \mathbb{C} \\
&= e^{A(\lambda)+B(\lambda)s} \prod_{\rho \in \mathcal{Z}_r} \Big(1 - \frac{s}{\rho}\Big) e^{\frac{s}{\rho}} \prod_{\rho \in \mathcal{Z}_c} \Big(1 - \frac{s}{\rho}\Big) e^{\frac{s}{\rho}} \\
&= e^{A(\lambda)+[B(\lambda)+c]s} \prod_{\rho \in \mathcal{Z}_r} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \prod_{i=1}^{\infty} \left(1 - \frac{s}{\rho_i}\right) \left(1 - \frac{s}{\bar{\rho}_i}\right) \\
&= e^{A(\lambda)+[B(\lambda)+c]s} \prod_{\rho \in \mathcal{Z}_r} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \prod_{i=1}^{\infty} \left(1 - \frac{s}{\rho_i}\right)^{m_i} \left(1 - \frac{s}{\bar{\rho}_i}\right)^{m_i}
\end{aligned} \tag{19}$$

Here λ denotes a mathematical object, $\varepsilon(\lambda)$ is a complex number of absolute value 1, $c = \sum_{i=1}^{\infty} \frac{2\alpha_i}{\alpha_i^2+\beta_i^2} < \infty$.

- $\rho_i = \alpha_i + j\beta_i$ and $\bar{\rho}_i = \alpha_i - j\beta_i$ ($\beta_i \neq 0$) are the complex-conjugate zeros of $\Lambda(\lambda, s)$, $a_n, n = 1, 2, \ldots, N$ are the real zeros of $\Lambda(\lambda, s)$;
- $m_i \geq 1$ is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, k - \rho_i, k - \bar{\rho}_i)$ of $\Lambda(\lambda, s)$;
- $0 \leq \alpha_i \leq k$, $0 < a_n < k$, $\sum_{i=1}^{\infty} \frac{m_i}{|\rho_i|^2} < \infty$;
- $\mathcal{Z}_r \cap \mathcal{Z}_c = \varnothing$.

Then

$$\begin{cases} \alpha_i = \frac{k}{2}, i \in \mathbb{N}, 0 < |\beta_1| < |\beta_2| < |\beta_3| < \cdots \\ a_n = \frac{k}{2}, n = 1 \end{cases} \tag{20}$$

i.e., all the zeros (both real and complex) of $\Lambda(\lambda, s)$ in the critical strip $0 \leq \Re(s) \leq k$ lie on the critical line $\Re(s) = \frac{k}{2}$.

Remark: Since the number of real zeros is finite and the complex zeros occur in conjugate pairs and satisfy $\sum_{i=1}^{\infty}\frac{m_i}{|\rho_i|^2}<\infty, 0\le\alpha_i\le k$, the decomposition $\prod_{\rho}=\left(\prod_{\rho\in\mathcal{Z}_r}\right)\left(\prod_{\rho\in\mathcal{Z}_c}\right)$ is valid; both factors converge (the first as a finite product, the second as an absolutely and uniformly convergent infinite product).

Proof. By Theorem 3, to determine the distribution of the complex zeros of $\Lambda(\lambda,s)$, it suffices to show that the newly introduced factors $\prod_{\rho\in\mathcal{Z}_r}\left(1-\frac{s}{\rho}\right)e^{\frac{s}{\rho}}$ and $\prod_{\rho\in\mathcal{Z}_r}\left(1-\frac{k-s}{\rho}\right)e^{\frac{k-s}{\rho}}$ do not affect the complex zeros related divisibility in the functional equation $\Lambda(\lambda,s)=\varepsilon(\lambda)\Lambda(\lambda,k-s)$. This follows from the given condition $\mathcal{Z}_r\cap\mathcal{Z}_c=\varnothing$, which implies that $\prod_{\rho\in\mathcal{Z}_r}\left(1-\frac{s}{\rho}\right)e^{\frac{s}{\rho}}$ and $\prod_{\rho\in\mathcal{Z}_c}\left(1-\frac{k-s}{\rho}\right)e^{\frac{k-s}{\rho}}$ are co-prime, $\prod_{\rho\in\mathcal{Z}_r}\left(1-\frac{k-s}{\rho}\right)e^{\frac{k-s}{\rho}}$ and $\prod_{\rho\in\mathcal{Z}_c}\left(1-\frac{s}{\rho}\right)e^{\frac{s}{\rho}}$ are co-prime, according to Ref.[3] (p.174, p.208) or Ref.[4] (see its THEOREM 4).

Thus, considering that the conditions in Theorem 4 cover those in Theorem 3, we conclude by Theorem 3 that

$$\Lambda(\lambda,s)=\varepsilon(\lambda)\Lambda(\lambda,k-s)\Rightarrow\begin{cases}\alpha_i=\frac{k}{2}, i\in\mathbb{N}\\ 0<|\beta_1|<|\beta_2|<|\beta_3|<\cdots\end{cases}\tag{21}$$

Next, we consider the real zeros of $\Lambda(\lambda,s)$.

By canceling the complex zeros related polynomial factors on both sides of $\Lambda(\lambda,s)=\varepsilon(\lambda)\Lambda(\lambda,k-s)$, we have

$$e^{A(\lambda)+[B(\lambda)+c]s}\prod_{n=1}^{N}\left(1-\frac{s}{a_n}\right)e^{\frac{s}{a_n}}=\varepsilon(\lambda)e^{A(\lambda)+[B(\lambda)+c](k-s)}\prod_{n=1}^{N}\left(1-\frac{k-s}{a_n}\right)e^{\frac{k-s}{a_n}}\tag{22}$$

where constant c is the same as in the proof of Theorem 3.

Further Eq.(22) is equivalent to

$$(-1)^N e^{A(\lambda)+[B(\lambda)+c']s}\prod_{n=1}^{N}\Big(s-a_n\Big)=\varepsilon(\lambda)e^{A(\lambda)+[B(\lambda)+c'](k-s)}\prod_{n=1}^{N}\Big(s-(k-a_n)\Big)\tag{23}$$

where $c'=c+\sum_{n=1}^{N}\frac{1}{a_n}$.

Suppose the multiplicity of zero $s = a_n$ is m_n ($m_n \geq 1$) that is finite and uniquely determined although unknown. Then Eq.(23) becomes

$$(-1)^N e^{A(\lambda)+[B(\lambda)+c']s} \prod_{t=1}^{T} \Big(s-a_t\Big)^{m_t} = \varepsilon(\lambda) e^{A(\lambda)+[B(\lambda)+c'](k-s)} \prod_{t=1}^{T} \Big(s-(k-a_t)\Big)^{m_t} \tag{24}$$

where $\sum_{t=1}^{T} m_t = N$.

Considering $(s - a_t)$ and $(s - (k - a_t))$ are irreducible in $\mathbb{R}[x]$, then by Lemma 1 or Lemma 2, Eq.(24) means

$$\begin{cases} (s - a_t) \mid (s - (k - a_l) \\ (s - (k - a_t)) \mid (s - a_l) \\ t, l = 1, 2, 3, \cdots, T \end{cases} \tag{25}$$

The only solution to Eq.(25) is $t = l, a_t = \frac{k}{2}, t = 1, \cdots, T$, otherwise the uniqueness of m_t would be violated with $a_t + a_l = k, t \neq l$. To avoid changing the multiplicity of m_t while $a_t = \frac{k}{2}$, we need to limit $T = 1$. Thus we get

$$\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\lambda, k - s) \Rightarrow a_t = \frac{k}{2}, t = 1, m_1 = N \tag{26}$$

Putting Eq.(21) and Eq.(26) together, we proved Eq.(20).

In addition to Eq.(20), we still have (by canceling the real zeros related polynomial factors on both sides of Eq.(24))

$$\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\lambda, k - s) \Longrightarrow (-1)^N e^{A(\lambda)+[B(\lambda)+c']s} = \varepsilon(\lambda) e^{A(\lambda)+[B(\lambda)+c'](k-s)} \tag{27}$$

That completes the proof of Theorem 4. □

We have the following supplementary remarks regarding Theorem 4.

1): If $\Lambda(\lambda, s)$ has poles/zeros at $s = 0, k$ with multiplicity (order) $r \geq 1$, then $s^r(k - s)^r\Lambda(\lambda, s)$ is an entire function. This adjustment does not affect the results of Theorem 4, since we have $s^r(k - s)^r\Lambda(\lambda, s) = \varepsilon(\lambda)(k - s)^r s^r \Lambda(\lambda, k - s) \Leftrightarrow \Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\lambda, k - s)$.

2): As pointed out in Ref.[5] (p.102), if $\rho_i = \alpha_i + j\beta_i$ is a zero of $\Lambda(\lambda, s)$, then $\bar{\rho}_i = \alpha_i - j\beta_i$ is a zero of $\Lambda(\overline{\lambda}, s)$, $\overline{\lambda}$ is the dual of λ. Therefore, to use Theorem 4 while $\lambda \neq \bar{\lambda}$ with symmetric functional equation $\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\overline{\lambda}, k - s)$, we need to construct a new symmetric functional equation

$\Lambda(\overline{\lambda}, s)\Lambda(\lambda, s) = \varepsilon(\lambda)\varepsilon(\overline{\lambda})\Lambda(\lambda, k-s)\Lambda(\overline{\lambda}, k-s)$ to ensure that the conjugate zeros appear together in the related Hadamard products.

3): Theorems 1-4, can not be applied to the Davenport-Heilbronn counterexample. That is because, in the proofs of Theorems 1-4, $\sum_{i=1}^{\infty} \frac{m_i}{|\beta_i|^2} < \infty, 0 \leq \alpha_i \leq k$ is a necessary condition. However, the Davenport-Heilbronn counterexample does not guarantee that all its zeros lie inside the critical strip since it possesses no Euler product (see Ref.[5] on page 102 for more details); therefore, one cannot deduce that $\sum_{i=1}^{\infty} \frac{m_i}{|\beta_i|^2} < \infty$ from $\sum_{i=1}^{\infty} \frac{m_i}{|\rho_i|^2} < \infty$ without $0 \leq \alpha_i \leq k$.

4 Main Results

We will make use of Theorem 4 to prove the extended Riemann Hypothesis (for Dedekind zeta functions), the generalized Riemann Hypothesis (for Dirichlet L-functions), and the grand Riemann Hypothesis (for cuspidal automorphic L-functions, and for any kind of L-functions, repectively).

In the following contents, the critical line means: $\Re(s) = \frac{1}{2}$, or more generally, $\Re(s) = \frac{k}{2}$, $k > 0$ is a real constant.

Ref.[5] (mainly in Section 5.1) provides a detailed account of the general properties of L-functions, including the unified definition (via Dirichlet series with Euler product), the gamma factor, the conductor, the root number, the completed L-function, the symmetric functional equation, the Hadamard product (as an entire function of order 1), among others.

In addition, we also have the following general property of L-functions, which was labeled Lemma 5.5 in Ref.[5] (p.101).

Lemma 5.5 [5]: Let $L(f, s)$ be an L-function. All zeros ρ of $\Lambda(f, s)$ are in the critical strip $0 \leq \sigma \leq 1$. For any $\epsilon > 0$, we have

$$\sum_{\rho \neq 0,1} |\rho|^{-1-\epsilon} < +\infty.$$

In Lemma 5.5, $\Lambda(f, s)$ is the completed L-function corresponding to $L(f, s)$, σ is the real part of ρ, and f is identical to λ in this paper as a symbol representing a mathematical object (e.g., Dirichlet character, modular form, automorphic representation).

Actually, in Lemma 5.5, the critical strip can be modified from the closed interval $0 \leq \sigma \leq 1$ to the open interval $0 < \sigma < 1$, according to Refs.[10-12].

To adapt to the wider situation of critical strip, i.e., $0 \leq \Re(s) \leq k$, Lemma 5.5 has been extended to Theorem 9 (positioned after Theorem 8).

Another general property of L-functions is as follows.

The zeros of the completed L-function $\Lambda(\lambda, s)$ are precisely the non-trivial zeros of $L(\lambda, s)$. See Ref.[5] (p.96) for more details (with f therein replaced by λ).

Thus, we can discuss the non-trivial zeros of L-functions based on the zeros of the corresponding completed L-functions.

4.1 Dirichlet L-Function

We briefly list the contents of Dirichlet L-functions relevant to this paper, based mainly on Ref.[5].

Definition: Let χ be a Dirichlet character modulo q. The Dirichlet L-function associated with χ is defined by

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}, \qquad \Re(s) > 1. \tag{28}$$

where the product is taken over all primes p.

For the principal (trivial) character χ_0 (where $\chi_0(n) = 1$ if $\gcd(n, q) = 1$ and $\chi_0(n) = 0$ otherwise), the L-function is related to the Riemann zeta function by:

$$L(\chi_0, s) = \zeta(s) \prod_{p|q} \left(1 - \frac{1}{p^s}\right) \tag{29}$$

Remark: The Riemann zeta-function $\zeta(s)$ is a special case of $L(\chi, s)$ with $\chi(n) \equiv 1$ ($q = 1$).

Completed L-function: Let χ be a primitive Dirichlet character modulo $q > 1$. The completed Dirichlet L-function is defined as:

$$\Lambda(\chi, s) = q^{\frac{s}{2}} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s+a}{2}\right) L(\chi, s) = \left(\frac{q}{\pi}\right)^{\frac{s}{2}} \Gamma\left(\frac{s+a}{2}\right) L(\chi, s) \tag{30}$$

where $a = 0$ if $\chi(-1) = 1$ (even character) and $a = 1$ if $\chi(-1) = -1$ (odd character).

Functional Equation: The completed Dirichlet L-function satisfies the functional equation:

$$\Lambda(\chi, s) = W(\chi)\Lambda(\overline{\chi}, 1-s) \tag{31}$$

where $W(\chi)$ is the root number:

$$W(\chi) = \frac{\tau(\chi)}{j^a\sqrt{q}} \tag{32}$$

with $j = \sqrt{-1}, \tau(\chi) = \sum_{n=1}^{q} \chi(n)e^{2\pi jn/q}$ being the Gauss sum associated with χ.

Hadamard Product: For primitive non-principal character χ, the completed L-function $\Lambda(\chi, s)$ is an entire function of order 1 and has the Hadamard product:

$$\Lambda(\chi, s) = e^{A(\chi)+B(\chi)s} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \tag{33}$$

where the product is over all zeros ρ of $\Lambda(\chi, s)$, and $A(\chi)$ and $B(\chi)$ are constants depending on χ.

For the principal character χ_0 modulo $q > 1$, Eq.(29) shows that the non-trivial zeros of $L(\chi_0, s)$ coincide with those of $\zeta(s)$, since the finite Euler product $\prod_{p|q}(1 - p^{-s})$ does not vanish in the critical strip; this case thus reduces to the case $q = 1$.

For $q = 1$, the trivial character χ_0 satisfies $\chi_0(n) \equiv 1$, so that $L(\chi_0, s) = \zeta(s)$ and, with $a = 0$,

$$\Lambda(\chi_0, s) = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

which carries simple poles at $s = 0$ and $s = 1$. Since the factor $s(1 - s)$ is invariant under $s \mapsto 1 - s$, the function

$$\Lambda'(\chi_0, s) = s(1-s)\Lambda(\chi_0, s)$$

is an entire function of order 1, and the functional equation $\Lambda(\chi_0, s) = \Lambda(\overline{\chi}_0, 1 - s)$ with $\overline{\chi}_0 = \chi_0$ and root number $W(\chi_0) = 1$ implies the symmetric functional equation

$$\Lambda'(\chi_0, s) = \Lambda'(\chi_0, 1-s).$$

Hence the Hadamard product applies to $\Lambda'(\chi_0, s)$. This is exactly the situation considered in the proof of Theorem 6, and it makes no difference to our result; thus this (trivial) case is omitted in the proof of Theorem 5.

Next we prove the generalized Riemann Hypothesis, noting that $0 < \Re(s) < 1$ is a subset of $0 \leq \Re(s) \leq 1$.

Theorem 5: The non-trivial zeros of Dirichlet L-functions in the critical strip $0 \leq \Re(s) \leq 1$ lie on the critical line.

Remark: The non-trivial zero problem of $L(\chi, s)$ for arbitrary non-principal characters reduces to that for primitive non-principal characters, so we only consider primitive non-principal characters in the following proof.

Remark: It suffices to prove that all the zeros of $\Lambda(\chi, s)$ in the critical strip $0 \leq \Re(s) \leq 1$ have real part $\frac{1}{2}$, i.e., all the zeros of $\Lambda(\chi, s)$ lie on the critical line.

Proof. We conduct the proof in two cases.

CASE 1: $\chi = \bar{\chi}$ (self-dual)

It suffices to verify that the properties of $\Lambda(\chi, s)$ match the following conditions of Theorem 4 with $\lambda = \chi$, $\chi = \bar{\chi}$, $\varepsilon(\lambda) = W(\chi)$, $k = 1$:

1) Symmetric functional equation;

2) Hadamard product expression;

3) The complex zeros of $\Lambda(\chi, s)$ appear in pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity $m_i \geq 1$. Furthermore m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$ of $\Lambda(\chi, s)$ under the symmetric functional equation;

4) $0 \leq \alpha_i \leq 1$, $\sum_{i=1}^{\infty} \frac{1}{|\rho_i|^2} < \infty$, $0 < a_n < 1, n \leq N$;

5) $\mathcal{Z}_r \cap \mathcal{Z}_c = \varnothing$.

Eq.(31) shows the symmetric functional equation; Hadamard product Eq.(33) is equivalent to Eq.(19) in Theorem 4 by separating all zeros into two sets $\mathcal{Z}_r$ and $\mathcal{Z}_c$; $\chi = \bar{\chi}$ guarantees that the complex conjugate zeros of $\Lambda(\chi, s)$ appear in pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity $m_i \geq 1$, and then m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$ by Eq.(31); The condition $0 \leq \alpha_i \leq 1$, $\sum_{i=1}^{\infty} \frac{1}{|\rho_i|^2} < \infty$, $0 < a_n < 1$ can be assured by Lemma 5.5, considering that $\sum_{i=1}^{\infty} \frac{1}{|\rho_i|^2}$ is a subseries of $\sum_{\rho \neq 0,1} \frac{1}{|\rho|^2}$; The condition $\mathcal{Z}_r \cap \mathcal{Z}_c = \varnothing$ holds because $\mathcal{Z}_r$ and $\mathcal{Z}_c$ are mutually exclusive sets, i.e., if $\rho \in \mathcal{Z}_r$, then $\Im(\rho) = 0$; if $\rho \in \mathcal{Z}_c$, then $\Im(\rho) \neq 0$.

Therefore, by Theorem 4 with $\lambda = \chi$, $\chi = \bar{\chi}$, $\varepsilon(\lambda) = W(\chi)$, $k = 1$, we know that all zeros (real, if any, and complex) of $\Lambda(\chi, s)$ in the critical strip

$0 \leq \Re(s) \leq 1$ lie on the critical line.

CASE 2: $\chi \neq \bar{\chi}$ (non-self-dual)

In this case, the complex conjugate zeros do not appear together in Eq.(33), because if ρ is a zero of $\Lambda(\chi, s)$, then $\bar{\rho}$ is a zero of $\Lambda(\bar{\chi}, s)$. To address this issue, we need to construct a new entire function $\Lambda(\chi, s)\Lambda(\bar{\chi}, s)$ with corresponding symmetric functional equation and Hadamard product expression.

First, we extend Eq.(31) to another form, i.e.,

$$\Lambda(\bar{\chi}, s) = W(\bar{\chi})\Lambda(\chi, 1-s) \tag{34}$$

Combining Eq.(34) with Eq.(31), we get a new symmetric functional equation

$$\Lambda(\bar{\chi}, s)\Lambda(\chi, s) = W(\bar{\chi})W(\chi)\Lambda(\chi, 1-s)\Lambda(\bar{\chi}, 1-s) \tag{35}$$

The product $\Lambda(\bar{\chi}, s)\Lambda(\chi, s)$ is an entire function of finite order ≤ 1, as this holds for the product of any two entire functions of order 1. Moreover, the zero set of $\Lambda(\bar{\chi}, s)\Lambda(\chi, s)$—being the union of the zeros of the individual factors—possesses the conjugate zeros paring property.

Then all complex zeros of $\Lambda(\bar{\chi}, s)\Lambda(\chi, s)$ come in conjugate pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity m_i; furthermore, by Eq.(35), m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$. In addition, Lemma 5.5 applies to $\Lambda(\bar{\chi}, s)\Lambda(\chi, s)$.

To be specific, based on Eq.(33), we have the following Hadamard product expression for $\Lambda(\chi, s)\Lambda(\bar{\chi}, s)$

$$\begin{aligned}
\Lambda(\chi, s)\Lambda(\bar{\chi}, s) &= \left[e^{A(\chi)+B(\chi)s} \prod_{\rho \in \mathcal{Z}_r} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \prod_{\rho \in \mathcal{Z}_c} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \right] \\
&\quad \cdot \left[e^{A(\bar{\chi})+B(\bar{\chi})s} \prod_{\bar{\rho} \in \mathcal{Z}_r} \left(1 - \frac{s}{\bar{\rho}}\right) e^{s/\bar{\rho}} \prod_{\bar{\rho} \in \mathcal{Z}_c} \left(1 - \frac{s}{\bar{\rho}}\right) e^{s/\bar{\rho}} \right] \\
&= e^{A(\chi)+A(\bar{\chi})+[B(\chi)+B(\bar{\chi})+c]s} \prod_{\rho \in \mathcal{Z}_r} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \prod_{i=1}^{\infty} \left(1 - \frac{s}{\rho_i}\right) \left(1 - \frac{s}{\bar{\rho}_i}\right)
\end{aligned} \tag{36}$$

where $c = \sum_{i=1}^{\infty} \frac{2\alpha_i}{\alpha_i^2 + \beta_i^2}$.

The other conditions required by Theorem 4 hold for the same reasons as in CASE 1.

Therefore, according to Theorem 4, all zeros (real, if any, and complex) of $\Lambda(\chi, s)\Lambda(\bar{\chi}, s)$, and consequently of $\Lambda(\chi, s)$, in the critical strip $0 \leq \Re(s) \leq 1$, lie on the critical line for $\chi \neq \bar{\chi}$.

Combining CASE 1 and CASE 2, we conclude that Theorem 5 holds as a specific case of Theorem 4 with $\lambda = \chi$, $\varepsilon(\lambda) = W(\chi)$, $k = 1$. □

Remark: According to Theorem 5: All non-trivial zeros (both real and complex) of Dirichlet L-functions, in the critical strip $0 \leq \Re(s) \leq 1$, lie on the critical line $\Re(s) = \frac{1}{2}$. This implies the non-existence of Landau-Siegel zeros. Hence, the Landau-Siegel zeros conjecture is justified.

4.2 Dedekind Zeta Function

We briefly list the contents of Dedekind zeta functions relevant to this paper, based mainly on Ref.[5].

Definition: For a number field $K/\mathbb{Q}$ of degree $d = [K : \mathbb{Q}]$ with ring of integers $\mathcal{O}_K$, the Dedekind zeta function is defined by

$$\zeta_K(s) = \sum_{\mathfrak{a} \neq 0} \frac{1}{(N\mathfrak{a})^s} = \prod_{\mathfrak{p}} \left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1}, \qquad \Re(s) > 1. \tag{37}$$

where the sum is over all nonzero integral ideals $\mathfrak{a}$ of $\mathcal{O}_K$, the product is over all nonzero prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$, and $N\mathfrak{a} = |\mathcal{O}_K/\mathfrak{a}|$ denotes the norm of the ideal $\mathfrak{a}$.

Remark: The Riemann zeta-function $\zeta(s)$ is a special case of $\zeta_K(s)$ with $K = \mathbb{Q}$, where $\mathbb{Q}$ denotes the field of rational numbers.

Completed Zeta Function: The completed Dedekind zeta function is defined as:

$$\Lambda_K(s) = |D_K|^{s/2} \pi^{-\frac{ds}{2}} \Gamma\left(\frac{s}{2}\right)^{r_1+r_2} \Gamma\left(\frac{s+1}{2}\right)^{r_2} \zeta_K(s) \tag{38}$$

where D_K is the discriminant of K, r_1 is the number of real embeddings of K, r_2 is the number of pairs of complex embeddings of K, $d = r_1 + 2r_2$.

Functional Equation: The completed Dedekind zeta function satisfies the functional equation

$$\Lambda_K(s) = \Lambda_K(1 - s) \tag{39}$$

Hadamard Product: The completed Dedekind zeta function has a simple pole at $s = 1$. The function $s(1-s)\Lambda_K(s)$ is an entire function of order 1 and has the Hadamard product:

$$s(1-s)\Lambda_K(s) = e^{A_K + B_K s} \prod_{\rho \neq 0,1} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \tag{40}$$

where the product runs over all zeros ρ of $\Lambda_K(s)$, and A_K and B_K are constants depending on K.

For more details of Dedekind zeta functions, please refer to Ref.[5] (Section 5.1 and Section 5.10) and Ref.[6] (Section 10.5.1).

Next, we prove the extended Riemann Hypothesis, noting that $0 < \Re(s) < 1$ is a subset of $0 \leq \Re(s) \leq 1$.

Theorem 6: The non-trivial zeros of Dedekind zeta functions in the critical strip $0 \leq \Re(s) \leq 1$ lie on the critical line.

Remark: It suffices to prove that all the zeros of $\Lambda_K(s)$ in the critical strip $0 \leq \Re(s) \leq 1$ have real part $\frac{1}{2}$, i.e., all the zeros of $\Lambda_K(s)$ lie on the critical line.

Proof. Let $\Lambda'_K(s) = s(1-s)\Lambda_K(s)$. Then $\Lambda'_K(s)$ is an entire function of order 1. And that, the zeros of $\Lambda_K(s)$ coincide with the zeros of $\Lambda'_K(s)$. Then, it suffices to show that the properties of $\Lambda'_K(s)$ match the conditions of Theorem 4 with $\lambda = \bar{\lambda} = K$, $\varepsilon(\lambda) = 1$, $k = 1$.

Actually, symmetric functional equation Eq.(39) guarantees that

$$\Lambda'_K(s) = s(1-s)\Lambda_K(s) = (1-s)s\Lambda_K(1-s) = \Lambda'_K(1-s) \tag{41}$$

Recalling Theorem 4, $\Lambda'_K(s)$ here corresponds to $\Lambda(\lambda, s)$, and Eq.(40) corresponds to Eq.(19).

Furthermore, the complex conjugate zeros of $\Lambda'_k(s)$ appear in pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity $m_i \geq 1$, since it is self-dual as $\Lambda_K(s)$ is, and then m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$ by Eq.(41); The other conditions required by Theorem 4 hold for the same reasons as in CASE 1 in the proof of Theorem 5.

Therefore, by Theorem 4, we know that all zeros (real, if any, and complex) of $\Lambda'_K(s)$ (thus of $\Lambda_K(s)$) in the critical strip $0 \leq \Re(s) \leq 1$ lie on the critical line. Thus, Theorem 6 holds as a specific case of Theorem 4 with $\lambda = \bar{\lambda} = K$, $\varepsilon(\lambda) = 1$, $k = 1$. □

4.3 Automorphic L-Function

We briefly list the contents of automorphic L-functions relevant to this paper, based mainly on Ref.[5].

Definition: Let $\pi \simeq \otimes'_v \pi_v$ be a cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$ with unitary central character, normalized such that the central character is trivial on the diagonally embedded copy of the positive reals, where $m \geq 1$ is an integer, $\simeq$ denotes isomorphism of representations, $\otimes'_v$ denotes the restricted tensor product over all places v of $\mathbb{Q}$ (including the archimedean place $v = \infty$), and $\mathbb{A}_{\mathbb{Q}}$ denotes the adele ring of $\mathbb{Q}$. In addition, Q_π is a positive integer, the arithmetic conductor of π.

The standard L-function associated to π is defined by

$$L(\pi, s) = \prod_{p<\infty} L_p(\pi, s) = \sum_{n=1}^{\infty} \frac{\lambda_\pi(n)}{n^s}, \qquad \Re(s) > 1. \tag{42}$$

where $L_p(\pi, s)$ is the local L-factor at the prime p. For each (finite) prime p, the inverse of the local factor $L_p(\pi, s)$ is a polynomial in p^{-s} of degree $\leq m$, i.e.

$$L_p(\pi, s) = \prod_{i=1}^{m} \left(1 - \alpha_{i,\pi}(p) p^{-s}\right)^{-1} \tag{43}$$

where $\alpha_{i,\pi}(p)$ are suitable complex numbers, $\alpha_{i,\pi}(p) \neq 0$ for all i whenever $p \nmid Q_\pi$ (π_p is unramified), and it might be the case that $\alpha_{i,\pi}(p) = 0$ for some i when $p \mid Q_\pi$ (π_p is ramified). Please refer to Ref.[9] for more details of coefficients $\alpha_{i,\pi}(p)$ and $\lambda_\pi(n)$.

At the archimedean place of $\mathbb{Q}$, there are m complex Langlands parameters $\mu_{i,\pi}$, based on which the archimedean factor $L_\infty(s, \pi)$ is defined by the following product of gamma factors

$$L_\infty(s, \pi) = \pi^{-\frac{ms}{2}} \prod_{i=1}^{m} \Gamma\left(\frac{s + \mu_{i,\pi}}{2}\right).$$

Remark: The Riemann zeta-function $\zeta(s)$ is a special case of $L(\pi, s)$ with $m = 1, \pi \simeq \mathbf{1}$ (the trivial representation).

Completed L-function: The completed automorphic L-function is defined by

$$\Lambda(\pi, s) = Q_\pi^{s/2} L_\infty(\pi, s) L(\pi, s) \tag{44}$$

Functional Equation: The completed automorphic L-function satisfies the functional equation

$$\Lambda(\pi, s) = \varepsilon(\pi)\, \Lambda(\tilde{\pi}, 1 - s) \tag{45}$$

where $\tilde{\pi}$ is the contragredient representation of π, $\varepsilon(\pi)$ is a complex number of absolute value 1, i.e., $|\varepsilon(\pi)| = 1$, also known as the root number of the corresponding L-function.

Hadamard Product: For cuspidal π on GL_m with $m \geq 2$, $\Lambda(\pi, s)$ is an entire function of order 1, and hence admits the Hadamard product expansion

$$\Lambda(\pi, s) = e^{A(\pi)+B(\pi)s} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \tag{46}$$

where the product runs over all zeros ρ of $\Lambda(\pi, s)$, and $A(\pi)$ and $B(\pi)$ are constants depending on π.

For cuspidal π on GL_1, $s^r(1-s)^r\Lambda(\pi, s)$, rather than $\Lambda(\pi, s)$ itself, is an entire function, $r \geq 1$ is the multiplicity (order) of pole of $\Lambda(\pi, s)$ at $s = 1$. This situation can be processed as shown in the proof of Theorem 6, it doesn't affect the final conclusion. Thus we ignore this situation in the proof of Theorem 7.

For more details of automorphic L-functions, please refer to Refs.[5][7][8].

The grand Riemann Hypothesis (GRH) has different versions of statements. To be specific, we adopt the following two versions.

GRH (version 1): For any cuspidal representation π, the non-trivial zeros of the L-function $L(\pi, s)$ in the critical strip $0 < \Re(s) < 1$ lie on the critical line $\Re(s) = \frac{1}{2}$.

GRH (version 2): The non-trivial zeros of any L-function in the critical strip $0 < \Re(s) < 1$ lie on the critical line $\Re(s) = \frac{1}{2}$.

GRH (version 1) corresponds to Theorem 7, GRH (version 2) corresponds to Theorem 8.

Theorem 7: The non-trivial zeros of cuspidal automorphic L-Functions in the critical strip $0 \leq \Re(s) \leq 1$ lie on the critical line.

Remark: It suffices to prove that all the zeros of $\Lambda(\pi, s)$ in the critical strip $0 \leq \Re(s) \leq 1$ have real part $\frac{1}{2}$, i.e., all the zeros of $\Lambda(\pi, s)$ lie on the critical line.

Proof. We conduct the proof in two cases.

CASE 1: $\pi \simeq \tilde{\pi}$ (self-dual, abbreviated as $\pi = \tilde{\pi}$)

It suffices to show that the properties of $\Lambda(\pi, s)$ match the conditions of Theorem 4 with $\lambda = \pi = \tilde{\pi} = \bar{\lambda}, k = 1$. Eq.(45) shows the symmetric functional equation; Hadamard product Eq.(46) is equivalent to Eq.(19) in Theorem 4 by separating all zeros into two sets $\mathcal{Z}_r$ and $\mathcal{Z}_c$; $\pi = \tilde{\pi}$ guarantees that the complex conjugate zeros of $\Lambda(\pi, s)$ appear in pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity $m_i \geq 1$, and then m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$ by Eq.(45); The condition $0 \leq \alpha_i \leq 1$, $\sum_{i=1}^{\infty} \frac{1}{|\rho_i|^2} < \infty$, $0 < a_n < 1$ can be assured by Lemma 5.5; The condition $\mathcal{Z}_r \cap \mathcal{Z}_c = \varnothing$ holds as explained in the proof of Theorem 5.

Therefore, by Theorem 4, we know that all zeros (real, if any, and complex) of $\Lambda(\pi, s)$ in the critical strip $0 \leq \Re(s) \leq 1$ lie on the critical line.

CASE 2: $\pi \not\simeq \tilde{\pi}$ (non-self-dual)

In this case, the complex conjugate zeros do not appear together in Eq.(46), because if ρ is a zero of $\Lambda(\pi, s)$, then $\bar{\rho}$ is a zero of $\Lambda(\tilde{\pi}, s)$. To address this issue, we need to construct a new entire function $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$ with corresponding symmetric functional equation and Hadamard product expression.

We first extend Eq.(45) to another form, i.e.,

$$\Lambda(\tilde{\pi}, s) = \varepsilon(\tilde{\pi})\Lambda(\pi, 1-s) \tag{47}$$

Combining Eq.(47) with Eq.(45), we get a new symmetric functional equation

$$\Lambda(\tilde{\pi}, s)\Lambda(\pi, s) = \varepsilon(\pi)\varepsilon(\tilde{\pi})\Lambda(\pi, 1-s)\Lambda(\tilde{\pi}, 1-s) \tag{48}$$

Both sides of Eq.(48) are the products of entire functions of order 1, thus they are still entire functions of order ≤ 1, as this holds for the product of any two entire functions of order 1. Moreover, the zero set of $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$—being the union of the zeros of the individual factors—possesses the conjugate zeros paring property.

Then all complex zeros of $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$ come in conjugate pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity m_i; furthermore, by Eq.(48), m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, 1-\rho_i, 1-\bar{\rho}_i)$. In addition, Lemma 5.5 applies to $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$.

Based on Eq.(46), we have the following Hadamard product expression for $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$

$$\Lambda(\tilde{\pi}, s)\Lambda(\pi, s) = e^{A(\pi)+A(\tilde{\pi})+[B(\pi)+B(\tilde{\pi})+c]s} \prod_{\rho \in \mathcal{Z}_r} \left(1-\frac{s}{\rho}\right) e^{\frac{s}{\rho}} \prod_{i=1}^{\infty} \left(1-\frac{s}{\rho_i}\right)\left(1-\frac{s}{\bar{\rho}_i}\right) \tag{49}$$

where $c = \sum_{i=1}^{\infty} \frac{2\alpha_i}{\alpha_i^2+\beta_i^2}$.

The other conditions required by Theorem 4 hold for the same reasons as in CASE 1.

Therefore, by Theorem 4, all zeros (real, if any, and complex) of $\Lambda(\tilde{\pi}, s)\Lambda(\pi, s)$, and consequently of $\Lambda(\pi, s)$, in the critical strip $0 \leq \Re(s) \leq 1$, lie on the critical line for $\pi \neq \tilde{\pi}$.

Combining CASE 1 and CASE 2, we conclude that Theorem 7 holds as a specific case of Theorem 4 with $\lambda = \pi, \bar{\lambda} = \tilde{\pi}, k = 1$. □

Actually, from the above proofs of Theorem 5, Theorem 6, and Theorem 7, we can note that each proof does not depend on the specific definition of the L-function $L(\lambda, s)$, but rather relies on the following general properties of $\Lambda(\lambda, s)$ and $L(\lambda, s)$:

P1: Symmetric functional equation between $\Lambda(\lambda, s)$ and $\Lambda(\bar{\lambda}, k-s)$: $\Lambda(\lambda, s) = \varepsilon(\lambda)\Lambda(\bar{\lambda}, k - s)$;

P2: $\Lambda(\lambda, s)$ or $s^r(k-s)^r\Lambda(\lambda, s)$ $(r \geq 1)$ is an entire function of order 1, with Hadamard product expression;

P3: The complex zeros in the relevant Hadamard products appear in pairs $(\rho_i, \bar{\rho}_i)$ with multiplicity $m_i \geq 1$;

P4: All zeros in the relevant Hadamard products lie in the critical strip $0 \leq \Re(\rho) \leq k$ (guaranteed by the Euler product property) and satisfy $\sum_{\rho \neq 0,k} \frac{1}{|\rho|^2} < \infty$;

P5: The disjointness of real and complex non-trivial zero sets;

P6: The zeros of $\Lambda(\lambda, s)$ are precisely the non-trivial zeros of $L(\lambda, s)$.

P1 and **P3** guarantee that m_i is the multiplicity of the i-th quadruple of zeros $(\rho_i, \bar{\rho}_i, k - \rho_i, k - \bar{\rho}_i)$.

Therefore, we have the following result on the non-trivial zero distribution of all kinds of L-functions.

Theorem 8: The non-trivial zeros of any L-function in the critical strip $0 \leq \Re(s) \leq k$ lie on the critical line if only the properties **P1**, **P2**, **P3**, **P4**, **P5**, and **P6** are satisfied.

Proof. It is not difficult to see that **P1**, **P2**, **P3**, **P4**, and **P5** cover all the conditions in Theorem 4. Thus, we know by Theorem 4 that all the zeros,

both real (if any) and complex, of $\Lambda(\lambda, s)$ in the critical strip $0 \leq \Re(s) \leq k$ lie on the critical line. Furthermore, according to **P6**, we conclude that all the non-trivial zeros, both real (if any) and complex, of $L(\lambda, s)$ in the critical strip $0 \leq \Re(s) \leq k$ lie on the critical line.
That completes the proof of Theorem 8. □

Remark: Conditions **P1-P3** and **P6** are well-known general properties of L-functions, see Chapter 5 of Ref.[5]. Condition **P4** follows from Theorem 9, which is an extended result of Lemma 5.5. In detail, $0 \leq \Re(\rho) \leq k$ reduces to the Euler product, while $\sum_{\rho \neq 0,k} \frac{1}{|\rho|^2} < \infty$ is a general result of complex analysis for entire functions of order 1. Condition **P5** is automatically satisfied by our definitions of $\mathcal{Z}_r$ (the set of zeros with $\Im(\rho) = 0$) and $\mathcal{Z}_c$ (the set of zeros with $\Im(\rho) \neq 0$). Therefore, Theorem 8 with conditions **P1-P6** is expected to serve as a guideline for constructing new zeta- or L-type functions.

Theorem 9: Let $L(\lambda, s)$ be an L-function, $\Lambda(\lambda, s)$, the corresponding completed L-function satisfying a functional equation of the form $\Lambda(\lambda, s) = \epsilon(\lambda)\Lambda(\bar{\lambda}, k - s)$, with $k \in \mathbb{R}, k > 0$. Then all zeros ρ of $\Lambda(\lambda, s)$ lie in the critical strip $0 \leq \Re(\rho) \leq k$. Moreover, for any $\epsilon > 0$, we have

$$\sum_{\rho \neq 0,k} |\rho|^{-1-\epsilon} < \infty.$$

where λ denotes a mathematical object, $\overline{\lambda}$ is the dual of λ; $\varepsilon(\lambda)$ is a complex number of absolute value 1, called the "root number" of L-function $L(\lambda, s)$.

Proof. Defining $s = kt$ and $F(\lambda, t) = \Lambda(\lambda, kt)$, the functional equation transforms as follows:

$$F(\lambda, t) = \Lambda(\lambda, kt) = \epsilon(\lambda)\Lambda(\bar{\lambda}, k - kt) = \epsilon(\lambda)\Lambda(\bar{\lambda}, k(1 - t)) = \epsilon(\lambda)F(\bar{\lambda}, 1 - t).$$

We thus obtain a new entire function $F(\lambda, t)$ with the standard functional equation $F(\lambda, t) = \epsilon(\lambda)F(\bar{\lambda}, 1 - t)$, to which Lemma 5.5 applies.

Let τ be a zero of $F(\lambda, t)$. By definition, $\rho = k\tau$ is then a zero of $\Lambda(\lambda, s)$, and this correspondence is bijective. By Lemma 5.5 we know that $0 \leq \Re(\tau) \leq 1$, $\sum_{\tau \neq 0,1} |\tau|^{-1-\epsilon} < \infty$ (for any $\epsilon > 0$), which immediately implies $0 \leq \Re(\rho) \leq k$, $\sum_{\rho \neq 0,k} |\rho|^{-1-\epsilon} < \infty$ (for any $\epsilon > 0$).

That completes the proof of Theorem 9. □

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